MARRIAGE OF TAYLOR AND FOURIER SERIES TO GENERATE DEFINITE INTEGRALS





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GOAL

- Our goal is to evaluate complicated definite integrals, as shown below, without using standard integration techniques (i.e. substitution; by parts…)

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta = ?$$

- In this presentation we will establish an unusual methodology for evaluating non-trivial definite integrals, some of which may or may not be new.

SOME PRIOR KNOWLEDGE: FOURIER SERIES

Below is the Definition of the Fourier series expansion of a function *f(x)*, [Churchill].

$$f(x) = \underbrace{a_0 + \sum_{n=1}^{\infty} a_n cos(nx)}_{cosine\ series} + \underbrace{\sum_{n=1}^{\infty} b_n sin(nx)}_{sine\ series}$$

For Even Functions

For Odd Functions

$$\begin{cases} a_0 = \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} f(x)\,dx \\ a_n = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \\ b_n = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \end{cases}$$

SOME PRIOR KNOWLEDGE: FOURIER SERIES

The Fourier Series allows us to express a non-periodic function as a summation of sinusoidal functions over the interval $(-\pi,\pi)$.

Example: Suppose we have the function $f(x) = x$. If we use the definition of the Fourier Series expansion, we will show that $f(x)$ has the following expansion:

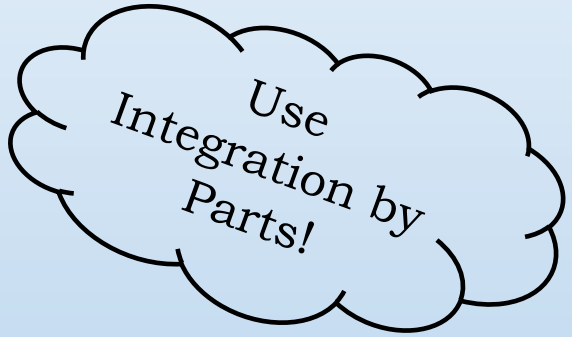
$$x = 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx)$$

We will first evaluate the coefficient integrals to obtain.

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\,dx = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos(nx)\,dx = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx = \frac{-2\cos n\pi}{n} = \frac{2(-1)^{n+1}}{n}$$

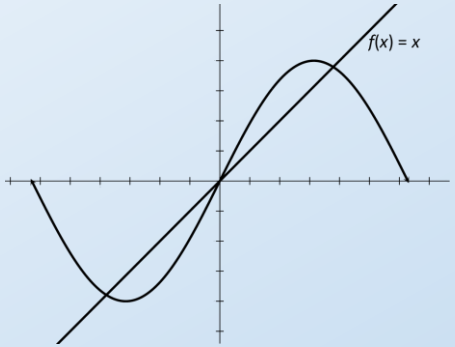


The first two integrals are 0, this allows us to express our function as a Sine Series:

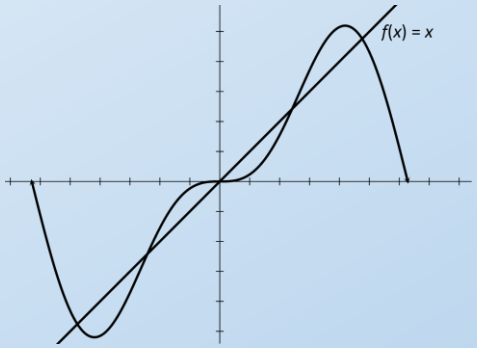
$$x = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n}\sin nx$$

Here are plots for the first 4 iterations of the Fourier Series expansion for $f(x) = x$.

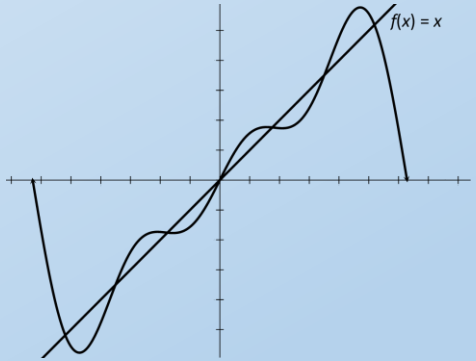
$$2\sum_{n=1}^{1}\frac{(-1)^{n+1}}{n}\sin(nx)$$



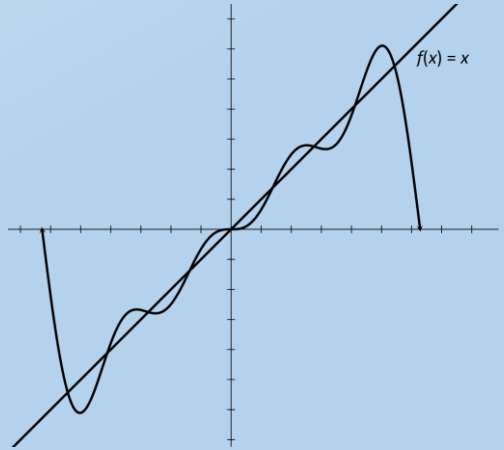
$$2\sum_{n=1}^{2}\frac{(-1)^{n+1}}{n}\sin(nx)$$



$$2\sum_{n=1}^{3}\frac{(-1)^{n+1}}{n}\sin(nx)$$



$$2\sum_{n=1}^{4}\frac{(-1)^{n+1}}{n}\sin(nx)$$



As we increase the number of iterations of the summation, we get better approximations to our function $f(x) = x$.

SOME PRIOR KNOWLEDGE: TAYLOR SERIES

Part of our methodology will involve using the Taylor Series expansion of complex functions. Recall the definition of the Taylor Series expansion of a function centered at $x = a$ [Stewart]:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Example: Suppose we let $f(x) = e^x$, then its Taylor series expansion centered at the origin is, $a = 0$:

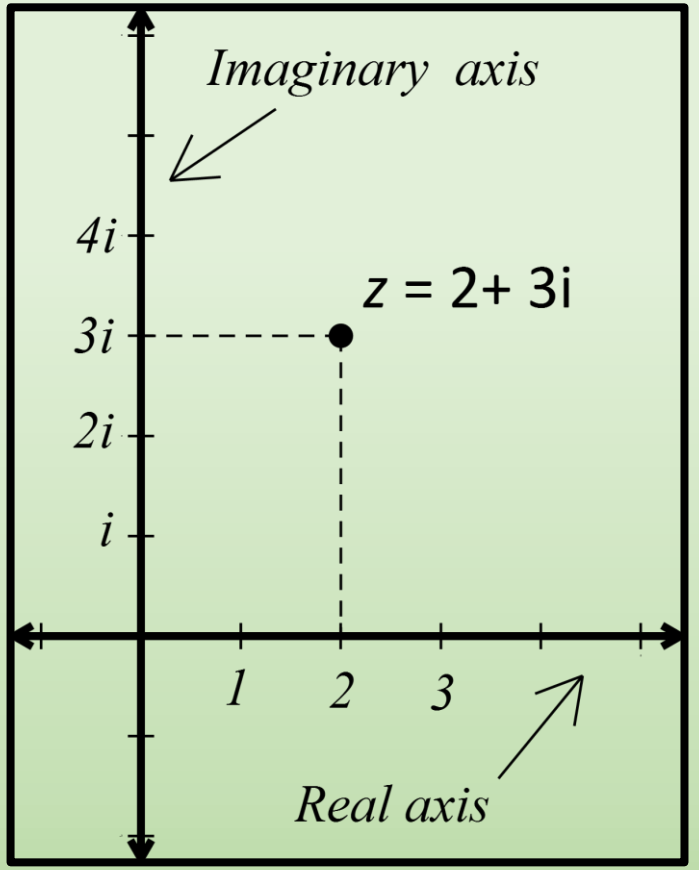
$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!}x^n$$

COMPLEX VARIABLES

Think of a complex variable as being a "2 dimensional variable". Complex numbers are represented by a real and imaginary portion:

$$z = \underbrace{x}_{REAL} + \overbrace{iy}^{IMAG.}$$

Example: Suppose we have $z = 2 + 3i$ we can graph this "2 dimensional" point in the complex plane, as shown to the right.



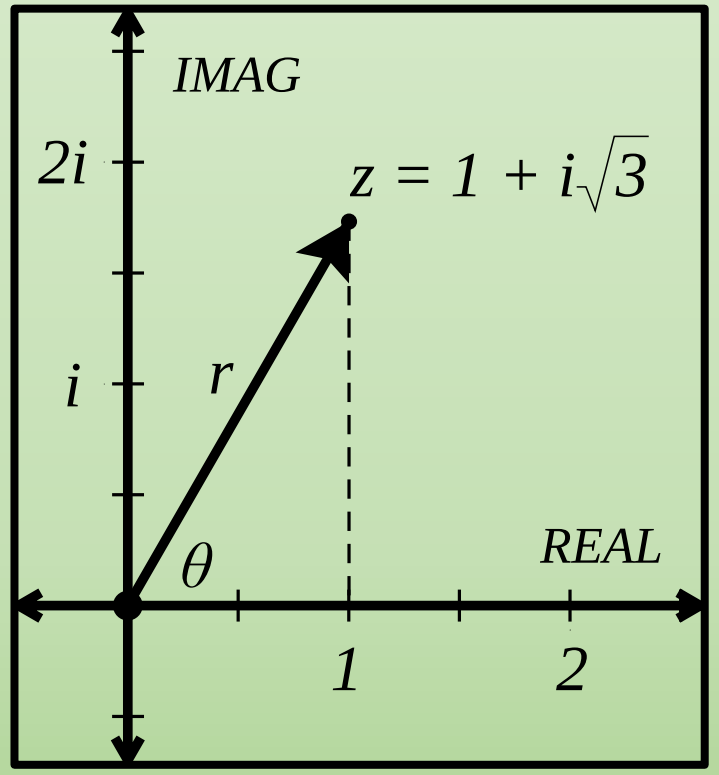
COMPLEX VARIABLES

A complex variable can also be viewed as a vector with magnitude "*r*" making an angle θ with the Real axis.

Example: Let us consider a familiar pair of numbers for our complex number:

$$z = 1 + i\sqrt{3}.$$

To the right is its graph. Using our high school trigonometry and geometry theorems we can conclude that $r = 2$ and $\theta = \pi/3$.



The x-component and y-component of our imaginary number can be expressed as:

$$x = 1 = 2\cos(\pi/3)$$

$$y = \sqrt{3} = 2\sin(\pi/3)$$

Therefore,

$$z = 1 + i\sqrt{3} = 2\cos(\pi/3) + i2\sin(\pi/3)$$

If we compare the line above with Euler's formula:

$$re^{i\theta} = r\cos\theta + ir\sin\theta$$

We can express our complex number in polar form:

$$z = 1 + i\sqrt{3} = 2e^{i\frac{\pi}{3}}$$

COMPLEX VARIABLES

From the previous example we can now express complex numbers in polar form:

$$z = x + iy = r\cos\theta + ir\sin\theta = re^{i\theta}$$

$$z = re^{i\theta}$$

Where, $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}\frac{y}{x}$. The above relation will be important to our methodology. Another important theorem that we will utilize is De Moivre's Theorem:

$$\left(re^{i\theta}\right)^n = r^n e^{in\theta} = r^n\cos n\theta + ir^n\sin n\theta$$

AN EXAMPLE

To demonstrate our methodology, let us begin with an example. Let $f(z) = e^z$, first we will decompose our function into real and imaginary components.

$$e^z = u(r,\theta) + iv(r,\theta)$$

By letting $z = re^{i\theta}$,

$$e^{\left(re^{i\theta}\right)} = e^{r\cos\theta + ir\sin\theta}$$

$$= e^{r\cos\theta} e^{ir\sin\theta}$$

$$\boxed{e^z = e^{r\cos\theta}\cos\left(r\sin\theta\right) + ie^{r\cos\theta} r\sin\left(r\sin\theta\right)} \quad (1)$$

$u(r,\theta)$ $\qquad$ $v(r,\theta)$

Let us now examine the Taylor Series expansion of our function:

$$e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

Let $z = re^{i\theta}$,

Use De Moivre's Theorem Here

$$e^{\left(re^{i\theta}\right)} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(re^{i\theta}\right)^n = \sum_{n=0}^{\infty} \frac{r^n}{n!} e^{in\theta} \therefore$$

$$\boxed{e^z = 1 + \sum_{n=1}^{\infty} \frac{r^n}{n!} \cos n\theta + i \sum_{n=1}^{\infty} \frac{r^n}{n!} \sin n\theta} \quad (2)$$

Observe that the line above closely resembles the Fourier Series.

Now we take the decomposition and the Taylor Series and we compare them with the Fourier Series Definition:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

② $$e^z = 1 + \sum_{n=1}^{\infty} \frac{r^n}{n!} \cos(n\theta) + i \sum_{n=1}^{\infty} \frac{r^n}{n!} \sin(n\theta)$$

① $$e^z = e^{r\cos\theta} \cos(r\sin\theta) + i e^{r\cos\theta} \sin(r\sin\theta)$$

This is the Cosine Series Expansion of u(r,θ)

This is the Sine Series Expansion of v(r,θ)

Notice from the previous slide we have found Fourier Series expansions, but more importantly we have stumbled upon solutions to definite integrals. Recall:

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx \quad \Rightarrow \quad 1 = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\,d\theta$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \quad \Rightarrow \quad \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \quad \Rightarrow \quad \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\sin(r\sin\theta)\sin(n\theta)\,d\theta$$

We have achieved our goal of evaluating complicated definite integrals. With some simple manipulation of the three integrals, and keeping '*r*' fixed and θ is the variable of integration we obtain:

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\,d\theta = 2\pi$$

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta = \frac{\pi r^n}{n!}$$

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\sin(r\sin\theta)\sin(n\theta)\,d\theta = \frac{\pi r^n}{n!}$$

This methodology works best if our original function can be conveniently decomposed to real and imaginary parts.

TABLES OF DEFINITE INTEGRALS

Below are some sample integrals that have been evaluated using this methodology.

(1) $$\int_{-\pi}^{\pi} \frac{a - r\cos\theta}{a^2 + r^2 - 2ar\cos\theta} d\theta = \frac{2\pi}{a}$$

(2) $$\int_{-\pi}^{\pi} \frac{a - r\cos\theta}{a^2 + r^2 - 2ar\cos\theta} \cos(n\theta) d\theta = \frac{\pi r^n}{a^{n+1}}$$

(3) $$\int_{-\pi}^{\pi} \frac{r\sin\theta}{a^2 + r^2 - 2ar\cos\theta} \sin(n\theta) d\theta = \frac{\pi r^n}{a^{n+1}}$$

(4) $$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta) d\theta = 2\pi$$

(5) $$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)\cos(2n\theta) d\theta = \frac{(-1)^n \pi r^{2n}}{(2n)!}$$

(6) $$\int_{-\pi}^{\pi} \sin(r\cos\theta)\sinh(r\sin\theta)\sin(2n\theta) d\theta = \frac{(-1)^{n+1} \pi r^{2n}}{(2n)!}$$

(34) $$\int_{-\pi}^{\pi} \cos\left(\frac{1}{2}\arctan\frac{r\sin\theta}{1 + r\cos\theta}\right) \sqrt[4]{1 + r^2 + 2r\cos\theta}\, d\theta = 2\pi$$

(35) $$\int_{-\pi}^{\pi} \cos\left(\frac{1}{2}\arctan\frac{r\sin\theta}{1 + r\cos\theta}\right) \sqrt[4]{1 + r^2 + 2r\cos\theta}\cos(n\theta) d\theta = \frac{\pi r^n \Gamma\left(\frac{3}{2}\right)}{\Gamma(n+1)\Gamma\left(\frac{3}{2} - n\right)}$$

(36) $$\int_{-\pi}^{\pi} \sin\left(\frac{1}{2}\arctan\frac{r\sin\theta}{1 + r\cos\theta}\right) \sqrt[4]{1 + r^2 + 2r\cos\theta}\sin(n\theta) d\theta = -\frac{\pi r^n \Gamma\left(\frac{3}{2}\right)}{\Gamma(n+1)\Gamma\left(\frac{3}{2} - n\right)}$$

THANK YOU

